

Difference between JPA, Hibernate and Spring Data JPA

1.Introduction

Persistence is the process of storing Java objects permanently in a database.

JPA, Hibernate, and Spring Data JPA are technologies used to simplify database operations in Java applications.

2. Java Persistence API (JPA)

- JPA stands for Java Persistence API.
- It is a specification (JSR 338) for persisting Java objects.
- It defines rules for Object Relational Mapping (ORM).
- It does not contain any implementation.
- Hibernate is one of the implementations of JPA.

3. Hibernate

- Hibernate is an ORM framework.
- It implements the JPA specification.
- It automatically converts Java objects into database tables.
- It reduces the amount of JDBC code required.
- It manages CRUD operations and transactions.

CODE:

```
/* Method to CREATE an employee in the database */
```

```
public Integer addEmployee(Employee employee){  
    Session session = factory.openSession();  
    Transaction tx = null;  
    Integer employeeID = null;  
    try {  
        tx = session.beginTransaction();  
        employeeID = (Integer) session.save(employee);  
        tx.commit();  
    } catch (HibernateException e) {  
        if (tx != null) tx.rollback();
```

```

        e.printStackTrace();
    } finally {
        session.close();
    }
    return employeeID;
}

```

4. Spring Data JPA

- Spring Data JPA is a Spring module.
- It is built on top of JPA.
- It does not implement JPA itself.
- It reduces boilerplate code.
- It provides ready-made CRUD methods such as save(), findAll(), delete(), etc.
- It manages transactions automatically using @Transactional.

CODE:

EmployeeRepository.java

```

public interface EmployeeRepository extends JpaRepository<Employee, Integer> {
}

```

EmployeeService.java

```

@Autowired
private EmployeeRepository employeeRepository;

@Transactional
public void addEmployee(Employee employee) {
    employeeRepository.save(employee);
}

```